

Question	Marking Guidance	Mark	Comments
01.1	1. Organic 2. Respiration 3. Carbon 4. Calorimetry;;	2 (2 x AO1)	4 correct = 2 marks 2–3 correct = 1 mark 0–1 correct = 0 marks 2. Ignore aerobic/anaerobic. 3. Accept cells/tissue(s)/plant(s)/animal(s)/organism(s)/NPP/ living material/biological molecules. 3. Ignore biomass/biological materials/organic matter and reference to 'dry'. 4. Accept calorimeter but reject colorimeter. 4. Accept phonetic spellings.
Question	Marking Guidance	Mark	Comments
01.2	1. Carbon dioxide combines/reacts with ribulose biphosphate/RuBP; 2. Produces two glycerate (3-)phosphate/GP using (enzyme) Rubisco; 3. GP reduced to triose phosphate; 4. Using reduced NADP; 5. Using <u>energy</u> from ATP; 6. Triose phosphate converted to glucose/hexose/RuBP/ribulose biphosphate/named organic substance;	6 (6 x AO1)	1 to 6. Accept marks in suitable diagram. 1. Accept idea of fixation for 'reacts'. 2. Accept: any answer which indicates that 2 x as much GP produced from one RuBP catalysed by Rubisco. 2 and 3. Reject GP once if incorrectly named e.g., glucose 3- phosphate. 3. Must have idea of reduction. This may be conveyed by stating mp4. 3 and 6. Only accept TP if triose phosphate is also in the answer. However only penalise once. 4. Accept NADPH or NADPH ₂ or NADPH + H for reduced NADP. 4. Reject: Any reference to reduced NAD for mp4 but allow reference to reduction for mp3. 4. and 5. must be in context of GP to triose phosphate.

